

MATH1141: Higher Mathematics 1B (Calculus)

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Chapter 1

Functions of several variables

1.1 Motivations, sketching surface, and partial derivatives

1.1.1 Function of two variables

We will consider functions of the type

$$z = F(x, y), \quad (x, y) \in D$$

where $D \subseteq \mathbb{R}^2$ is the domain of F and F is real-valued. We write $F : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$.

For example:

- $F(x, y) = 2x + y^6$ and $D = \mathbb{R}^2$
- $F(x, y) = xe^y$ and $D = \mathbb{R}^2$

Note: if $D = \mathbb{R}^2$, sometimes the domain is not explicitly written.

Geometric representation

The function of one variable: $y = f(x)$, $x \in [a, b]$ can be represented as a curve in $\mathbb{R}^2$.

Similarly, a function of two variables: $z = F(x, y)$, $(x, y) \in D$ can be described as a surface in $\mathbb{R}^3$

1.1.2 Sketching and visualising surfaces

Consider a surface defined by

$$z = F(x, y).$$

Conventionally, z is the distance above or below the x, y -plane.

A **level curve** of a function, $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, is a curve in $\mathbb{R}^2$ defined by $F(x, y) = C$, where C is a constant. This can be visualised by cutting a surface with a plane. Typically this plane is parallel to the xy -plane

Example

Sketch the surface defined by

$$z = F(x, y) = x^2 + y^2$$

1. Sketch some level curves for different values C .
 - Set $z = C = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow$ Singular point at origin
 - Set $z = C = 1 \Rightarrow x^2 + y^2 = 1 \Rightarrow$ Circle centred at 0 with radius 1
 - Set $z = C = 4 \Rightarrow x^2 + y^2 = 4 \Rightarrow$ Circle centred at 0 with radius 2
 - Set $z = C = 8 \Rightarrow x^2 + y^2 = 8 \Rightarrow$ Circle centred at 0 with radius $2\sqrt{2}$
2. Sketch the curves of intersection between the surface $z = x^2 + y^2$ and the yz -plane (set $x = 0$) or xz -plane (set $y = 0$).
 - In this example, both the intersections form parabolas.

Example

Sketch the surface defined by $x^2 + y^2 - z^2 = 1$.

1. Level curves
 - level curves with $z = 0$ is a circle with radius 1
 - level curves with $z = \pm 1$ are circles with radius $\sqrt{2}$
 - level curves with $z = \pm 2$ are circles with radius $\sqrt{5}$
2. Intersections between the surface and the yz -plane gives $y^2 - z^2 = 1$.
3. These pieces of information can be put together to form the surface

1.1.3 Partial differentiation

Informally: differentiate with respect to one of the variables by holding the other variable fixed.

Definition

If $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function of two variables x and y , then the partial derivatives of $F(x, y)$ with respect to x and y at the point (x_0, y_0) are defined by

$$\frac{\partial F}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h},$$

and

$$\frac{\partial F}{\partial y}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{F(x_0, y_0 + h) - F(x_0, y_0)}{h},$$

wherever these limits exist.

If a function F has partial derivatives with respect to x and y , then:

- The partial derivative with respect to x may be denoted by

$$\frac{\partial F}{\partial x} \equiv F_x \equiv D_1 F.$$

- The partial derivative with respect to y may be denoted by

$$\frac{\partial F}{\partial y} \equiv F_y \equiv D_2 F.$$

The rules for partial derivatives are similar to that of normal derivatives.

Example

Consider

$$F(x, y) = e^{y^2} \cos(x^2 y) + e^{x^2 y} \sin(y).$$

Compute the partial derivatives of $\frac{\partial F}{\partial x}$ and $\frac{\partial F}{\partial y}$.